MEETSYNC_JOURNAL

Project Title: MeetSync – Professor–Student Meeting Scheduler
Course: UCS503 – Software Engineering Lab
Institution: Thapar Institute of Engineering & Technology, Patiala
Department: Computer Science and Engineering
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Project Team

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Week 1: Project Ideation and Selection

Objectives

- Identify a practical problem that can be addressed through software.
- Explore possible project ideas.
- Finalize the project scope and objectives.

Work Completed

During the first week, the team discussed various software project ideas and evaluated them based on their usefulness, feasibility, complexity, and relevance to the Software Engineering Lab.

We identified the difficulties students and professors face while scheduling academic consultations. The traditional approach of communicating through emails or informal messages can result in delayed responses, unclear availability, and overlapping appointments.

Based on these observations, we selected **MeetSync – Professor–Student Meeting Scheduler** as our project. We defined the initial scope, including availability management, meeting requests, approval and rejection, rescheduling, cancellations, and notifications.

We also identified the primary users of the system: students, professors, and administrators.

Division of Work

- **Yogesh Goyal:** Evaluated the project idea, identified the problem statement, and outlined the proposed solution.
- **Rakshit Chaudhary:** Researched the existing scheduling process and identified its limitations.
- **Tanish Kumar Singla:** Helped evaluate project feasibility, define objectives, and establish the initial feature list.

The division above is a suggested allocation and should be updated to reflect the actual contributions of each member.

Challenges Faced

The initial challenge was selecting a project that was useful yet manageable within the available academic timeline. We also needed to avoid choosing a topic that was either too basic or too complex.

How We Addressed Them

We evaluated the project in terms of its core functionality and divided the idea into manageable modules. This helped us establish a realistic initial scope while leaving room for future enhancements.

Outcome

The project idea, problem statement, primary users, and initial objectives were established.

Week 2: Requirement Analysis and Project Planning

Objectives

- Understand the requirements of students, professors, and administrators.
- Identify functional and non-functional requirements.
- Select a suitable Software Development Life Cycle model.

Work Completed

During the second week, we analyzed the proposed system in greater detail. We identified the tasks each user should be able to perform and converted them into system requirements.

For students, the requirements included viewing professor availability, requesting meetings, tracking meeting status, and cancelling eligible appointments.

For professors, the requirements included managing availability, viewing requests, accepting or rejecting meetings, and rescheduling appointments.

We also identified administrative requirements such as system configuration and report generation.

The team selected **Agile Scrum** as the proposed SDLC model because it supports incremental development, continuous feedback, and iterative testing. The project was divided into modules such as authentication, availability management, meeting scheduling, and notifications.

Division of Work

- **Yogesh Goyal:** Drafted the problem statement, project objectives, and proposed solution.
- **Rakshit Chaudhary:** Identified functional requirements and the main interactions between users and the system.
- **Tanish Kumar Singla:** Worked on non-functional requirements, project feasibility, and the justification for selecting Agile Scrum.

Challenges Faced

One challenge was distinguishing functional requirements from non-functional requirements. Another was deciding which features were essential to the first version and which could be treated as optional enhancements.

How We Addressed Them

We classified requirements according to their purpose. Features describing system actions were listed as functional requirements, while performance, security, reliability, and usability were documented as non-functional requirements.

We prioritized authentication, availability management, and meeting booking as core capabilities.

Outcome

The project scope, user requirements, initial feature priorities, and development methodology were defined.

Week 3: Use-Case Modeling and User Interaction Analysis

Objectives

- Identify actors and their interactions with the system.

- Prepare use-case diagrams.
- Document the main use cases and their execution flows.

Work Completed

During the third week, we developed the use-case model for MeetSync.

We identified Student, Professor, and Administrator as the primary actors. We also considered the notification system and an optional payment gateway in the expanded system design.

The team mapped the major use cases, including registration and login, viewing availability, requesting meetings, managing availability, accepting or rejecting requests, rescheduling, cancellation, and viewing meeting history.

We prepared use-case templates containing descriptions, preconditions, main flows, alternative flows, and postconditions.

We also examined relationships between use cases, including authentication, availability checking, and notification generation.

Division of Work

- **Yogesh Goyal:** Worked on identifying actors, student-related use cases, and meeting request scenarios.
- **Rakshit Chaudhary:** Worked on professor-related use cases, including availability management and request processing.
- **Tanish Kumar Singla:** Worked on administrative use cases, use-case relationships, and documentation of detailed use-case templates.

Challenges Faced

The main challenge was representing all important interactions without making the use-case diagram overly complicated. We also needed to distinguish independent use cases from actions that were included in or extended another use case.

How We Addressed Them

We grouped related activities by actor and reviewed the diagram against the defined requirements. We documented detailed execution steps separately in use-case templates to keep the diagram readable.

Outcome

The use-case diagram and detailed use-case templates were prepared to describe the expected behavior of the system.

Week 4: Activity Diagram and Swimlane Modeling

Objectives

- Represent the complete meeting scheduling workflow.
- Identify decision points and alternative paths.
- Clarify the responsibilities of each participant.

Work Completed

During the fourth week, we prepared the activity diagram representing the workflow from user authentication to meeting completion.

The workflow included opening the application, logging in, viewing professor availability, selecting a time slot, checking availability, creating a meeting request, processing the request, updating its status, and sending notifications.

We incorporated decision points for invalid credentials and unavailable meeting slots.

We also prepared a swimlane diagram to separate the responsibilities of the Student, System, Professor, and Notification System.

The swimlane representation helped us visualize which activities are performed by users and which are handled automatically by the system.

Division of Work

- **Yogesh Goyal:** Worked on the student-side workflow, including availability search and meeting requests.
- **Rakshit Chaudhary:** Worked on the professor-side workflow, including approvals, rejections, and rescheduling.
- **Tanish Kumar Singla:** Worked on system activities, notification flows, and the overall swimlane representation.

Challenges Faced

A major challenge was maintaining consistency between the use-case diagram and the activity diagram. We also needed to represent alternative outcomes without creating confusing or unnecessarily long flows.

How We Addressed Them

We traced the workflow from start to finish and checked that each major action corresponded to a previously identified requirement. Decision branches were added for important failure conditions, such as invalid credentials and unavailable slots.

Outcome

The activity diagram and swimlane diagram were prepared to describe the workflow and the responsibilities of different participants.

Week 5: Data Flow Diagram Development

Objectives

- Understand how data moves through the proposed system.
- Prepare DFDs at different levels of abstraction.
- Identify major processes and data stores.

Work Completed

During the fifth week, we focused on Data Flow Diagrams (DFDs).

We prepared a Level 0 DFD representing MeetSync as a single process interacting with students and professors. This established the system boundary and identified the main information exchanged with external entities.

Next, we developed the Level 1 DFD by decomposing the system into four major processes:

1. User Authentication and Profile Management.
2. Faculty Availability Management.
3. Meeting Request and Scheduling Engine.
4. Notification and Alert Dispatcher.

We also documented the corresponding data stores for users, availability, meeting records, and notifications.

Finally, we prepared the Level 2 DFD to illustrate the decomposition of processes into smaller activities, such as credential validation, session creation, availability checking, meeting creation, and notification generation.

Division of Work

- **Yogesh Goyal:** Worked on the context-level DFD and student-related data flows.
- **Rakshit Chaudhary:** Worked on the meeting scheduling process and faculty availability data flows.
- **Tanish Kumar Singla:** Worked on authentication, notification processes, data stores, and the detailed DFD documentation.

Challenges Faced

The main difficulty was deciding the correct level of detail for each DFD. We needed to avoid including too many internal details in the Level 0 diagram while ensuring that the Level 1 and Level 2 diagrams provided sufficient decomposition.

Another challenge was maintaining consistency in process names, data flows, and data stores across the different levels.

How We Addressed Them

We developed the diagrams progressively, starting with the overall system context before decomposing major processes. We reviewed the inputs and outputs of each process to ensure that the diagrams remained consistent with the requirements and workflow.

Outcome

The Level 0, Level 1, and Level 2 DFDs were prepared to explain the system boundary, major processes, data movement, and detailed process decomposition.

Week 6: Software Requirements Specification

Objectives

- Consolidate the system requirements into a formal SRS.
- Document interfaces, constraints, and quality requirements.
- Examine the technical, operational, and economic feasibility of the project.

Work Completed

During the sixth week, we prepared the Software Requirements Specification (SRS) using an IEEE-style structure.

The document included the purpose and scope of MeetSync, product perspective, user classes, functional requirements, non-functional requirements, external interfaces, constraints, and feasibility analysis.

We documented functional requirements for authentication, availability scheduling, booking and conflict checking, approval management, cancellation, status tracking, profiles, notifications, reporting, and system configuration.

We also recorded proposed quality targets relating to performance, security, reliability, usability, and data integrity.

The proposed technology stack was documented as React for the frontend, Node.js for backend services, PostgreSQL for the database, bcrypt for password hashing, and JWT for authentication.

The feasibility study considered whether the project could be developed using commonly available technologies and standard development resources.

Division of Work

- **Yogesh Goyal:** Worked on the SRS introduction, purpose, scope, and functional requirements.
- **Rakshit Chaudhary:** Worked on non-functional requirements, security considerations, and external interface requirements.
- **Tanish Kumar Singla:** Worked on constraints, feasibility analysis, technology selection, and consistency review.

Challenges Faced

The primary challenge was converting the earlier diagrams and discussions into precise, unambiguous requirements. We also needed to ensure that the proposed security and performance expectations were clearly documented without confusing design targets with tested results.

How We Addressed Them

We organized requirements into categories and assigned unique identifiers to functional requirements. We reviewed the SRS against the use cases and DFDs to identify inconsistencies and missing requirements.

We treated performance and reliability values as proposed targets rather than measured achievements.

Outcome

The SRS documented the expected system behavior, quality requirements, interfaces, constraints, and feasibility considerations.

Week 7: User Stories, Story Cards, and Requirement Validation

Objectives

- Express system requirements from the perspective of end users.
- Define acceptance criteria for important features.
- Check consistency across the project documentation.

Work Completed

During the seventh week, we developed user stories and story cards for the main functionalities of MeetSync.

The user stories described what students, professors, and administrators needed from the system and why those capabilities were useful.

Examples included viewing professor availability, requesting meetings, viewing meeting status, cancelling appointments, managing availability, processing meeting requests, rescheduling meetings, receiving notifications, and generating reports.

We prepared acceptance criteria for these stories. For example, a meeting request should be checked against availability, scheduling conflicts should be prevented, and a request should be created only when the selected slot is valid.

We also reviewed the use-case diagrams, activity diagram, DFDs, and SRS to check that the requirements were represented consistently.

Division of Work

- **Yogesh Goyal:** Worked on student-focused user stories, including availability viewing, meeting requests, and cancellation.
- **Rakshit Chaudhary:** Worked on professor-focused user stories, including availability management, approvals, and rescheduling.
- **Tanish Kumar Singla:** Worked on notification and administration stories, story-card formatting, and documentation review.

Challenges Faced

One challenge was ensuring that user stories were specific enough to be useful. Broad statements did not always make it clear how the functionality would be evaluated.

We also needed to maintain consistency between acceptance criteria and the detailed use-case flows.

How We Addressed Them

We used the standard user-story structure: “As a [user], I want [functionality], so that [benefit].” We then added concrete acceptance criteria describing the expected outcomes.

We cross-checked the stories against the existing requirements and diagrams to reduce duplication and identify gaps.

Outcome

The user stories and story cards were prepared, and the requirements documentation was reviewed for consistency.

Week 8: Documentation Consolidation and Mid-Semester Evaluation

Objectives

- Consolidate the work completed during the previous seven weeks.
- Review the project report for completeness and consistency.
- Prepare the documented work for evaluation.

Work Completed

During the eighth week, we focused on consolidating the project documentation into a structured Software Engineering project report.

We organized the report into the Project Selection Phase and Analysis Phase. The report included the software bid, project overview, use-case diagrams and templates, activity and swimlane diagrams, Level 0 to Level 2 DFDs, the SRS, user stories, and story cards.

We reviewed the diagrams and accompanying explanations to ensure that they represented the intended system functionality.

We also reviewed the project objectives and requirements to make sure that the report communicated the purpose, scope, proposed behavior, and expected quality of MeetSync.

The team prepared the report for the mid-semester evaluation and identified areas that could be addressed during subsequent development work.

Division of Work

- **Yogesh Goyal:** Worked on consolidating the project overview, objectives, and relevant requirements documentation.
- **Rakshit Chaudhary:** Reviewed the diagrams, scheduling workflows, and functional descriptions.
- **Tanish Kumar Singla:** Worked on report formatting, documentation organization, and final consistency checks.

Challenges Faced

The main challenge was bringing together multiple diagrams and written sections while maintaining consistent terminology and avoiding contradictions.

Another challenge was clearly distinguishing the design and requirements completed so far from implementation work that may still be pending.

How We Addressed Them

We reviewed the report section by section and checked the diagrams against the written descriptions. We organized the documentation according to the required project phases and ensured that the report clearly described the proposed design rather than claiming unverified implementation or testing results.

Outcome

The work completed during the eight-week period was consolidated into the project report for mid-semester evaluation. The project had established its requirements, major system workflows, data flow models, SRS, and user stories.
